METRIC

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SUPERSEDING
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DEPARTMENT OF DEFENSE
STANDARD PRACTICE

CADMIUM PLATING, LOW EMBRITTLEMENT, ELECTRODEPOSITION

AMSC: N/A

AREA: MFFP

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FOREWORD

This military standard is approved for use by AF-70 Hill AF Base, AFSC Hill 309 CMXG, Department of the Air Force, and is available for use by all departments and agencies of the Department of Defense.

This standard provides guidance for the Air Force repair process, acquisition, and manufacture of parts and/or spare parts on the landing gear of all military aircraft.

Comments, suggestions, or questions on this document should be addressed to: AFMC 417 SCMS/GUEA, Hill AFB, UT 84056-5228 or email to: 417SCMS.GUEA.Workflow@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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SUMMARY OF REVISION D MODIFICATIONS

1. Changed the Uniform Resource Locator (URL) for the ASSIST Online data base throughout the document.

2. Changed MIL-STD header on each page to reflect document's change status

3. Formatted paragraphs, sentences, tables to a standard consistency

4. Front page:
   Added new DoD seal
   Deleted "Note" box
   Added Distribution A statement

5. Page ii:
   Changed POC program name
   Changed verbiage IAW MIL-STD-962D

6. Page iii – iv: Added new page "SUMMARY OF REVISION MODIFICATIONS"

7. Page v – vi: "Contents" changed numerous page numbers to correspond with modifications

8. Page 1:

9. Page 2:

10. Page 3:
    Changed Materials statement.
    Added Toxic chemical, hazardous substance and ozone-depleting chemical statement.
    Deleted all standard references.

11. Page 4:

12. Page 8:

13. Page 12:

14. Page 13:

15. Page 14:

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1. SCOPE

1.1 Scope. This standard covers the process and materials required for the electro-deposition of cadmium on high strength steel substrates. Subsequent heat treating techniques needed to insure low embrittlement of steel are also described. This process is not authorized for use on steel heat treated above 1.65×10^9 PASCAL's (Pa) (240,000 pounds per square inch {psi}) without the approval of the applicable procuring activity.

1.2 Documentation. This standard meets the corrosion resistance requirements of SAE-AMS-QQ-P-416. It may be used whenever SAE-AMS-QQ-P-416 is specified provided any unique requirements in SAE-AMS-QQ-P-416 are compiled with.

1.3 Classification.

1.3.1 Classes. If no class is specified in the manual or drawing, Class 2 should be used. For surfaces that cannot be touched by a 0.75 inch sphere, including internal threads, no plating thickness is required, but such areas must show evidence of plating. If not specified, the maximum for each Class will be the minimum plus 0.0003 inch, except in high cathode current density areas such as corners and edges. Cadmium plating covered by this standard identifies the following Classes:

a.	Class 1	0.013mm (0.0005 inch) thick minimum
b.	Class 2	0.008mm (0.0003 inch) thick minimum
c.	Class 3	0.005mm (0.0002 inch) thick minimum

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 4 and 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements or documents cited in sections 4 and 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, handbooks, and commercial item descriptions. The following specifications, standards, handbooks, and commercial item descriptions form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5002	Surface Treatments and Inorganic Coatings for Metal Surfaces of Weapon System

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-871	Electro-chemical Stripping of Inorganic Finishes
MIL-STD-1504	Abrasive Blasting of Aircraft Components

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COMMERCIAL ITEM DESCRIPTIONS (CID)

A-A-50800	Cadmium Oxide
A-A-51126	Anodes, Cadmium

(Copies of these documents are available online at https://assist.dla.mil)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

29 CFR 1910	Toxic and Hazardous Substances

(Copies of this document are available online at https://www.osha.gov/)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM F 519	Mechanical Hydrogen Embrittlement Evaluation of Plating Processes & Service Environments

(Copies of this document are available online at www.astm.org.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE) INTERNATIONAL

SAE-AMS-QQ-P-416	Plating, Cadmium (Electrodeposited)

(Copies of this document are available online at https://www.sae.org)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document; however, supersedes applicable law and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 High strength steel. For the purpose of this standard, high strength steel is defined as steel heat treated to 1.24×10^{9} Pa (1.26×10^{2} kgf/mm) (1.8×10^{5} psi) and above.

3.2 Material batch. All items processed at one time through the plating bath.

4. GENERAL REQUIREMENTS

4.1 General. This section covers the materials and equipment required to accomplish the described process.

4.2 Materials and Equipment.

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4.2.1 Materials. The use of reclaimed materials shall be encouraged to the maximum extent possible. Materials used in cadmium plating are as follows:

a. Sodium Hydroxide, flake, granulated, or liquid form.

b. Sodium Cyanide, plating grade (96 – 98% NaCN).

c. Cadmium Ball Anodes (A-A-51126)

d. Cadmium Oxide (A-A-50800).

4.2.2 Equipment. Equipment used in cadmium plating is as follows:

a. Either generated or rectified D.C. current may be used. Ripple value shall not exceed 10 percent as measured by dividing the Root Mean Square of the A.C. voltage component by the D.C. voltage.

b. Tanks shall be resistant to the operating temperature and the chemical environment. Tanks, in which any electrolytic action is taking place, must be free of shorts.

c. Processing tanks to be operated at temperatures other than room temperature shall be equipped with automatic temperature indicating and regulating devices.

d. An ammeter shall be placed in series with the cadmium tank cathode. The ammeter shall have sufficient shunts and switches to provide a full-scale reading equal to the maximum capacity of the power source, and an accuracy of ±10 percent of the current being measured.

e. A blast cabinet shall be located near the plating line. The size of the cabinet shall be adequate to enclose the parts to be plated. Air lines shall be suitably trapped and filtered to prevent in-process contamination of the parts to be cleaned.

f. An oven capable of baking parts at 190.5 ± 13.9°C (375 ± 25°F) shall be located near the plating line. The size of the oven shall be adequate to enclose parts to be plated. The oven shall be equipped with temperature indication, recording, and regulating devices.

4.3 Specification SAE-AMS-QQ-P-416. The requirements of SAE-AMS-QQ-P- 416 shall be compiled with on all parts, in conjunction with those specified in this standard. If there is a conflict between the two documents; however, the requirements of this standard shall be governed.

4.4 Embrittlement. Qualification test specimens and process control test specimens shall be subjected to a sustained load test at 75 percent of the ultimate notched tensile strength. The specimens shall endure this sustained load for 200 hours minimum without failing or cracking.

4.5 Finish. The plated part will have a finish that is smooth, continuous, homogeneous, adherent, and free from pits, blisters, nodules, and any other indications of harmful defects. The appearance of a properly applied plate may vary from a dull grey to a frosty white. A bright, dense deposit indicates malfunction of the process, which may produce embrittled parts (See: MIL-S-5002).

4.6 Re-processing. Parts rejected for defective plating, requiring stripping and re-plating, shall include all of the pre-plating steps of this standard. Parts shall be stripped in accordance with MIL-STD-871.

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5. DETAILED REQUIREMENTS

5.1 General. This section provides the detailed requirements required to accomplish the plating process identified in this standard.

5.2 Prior to plating. Prior to plating, all machining, forming, welding, and shot peening shall be completed.

5.2.1 Baking. Parts shall be baked for stress relief before plating for four (4) hours minimum at 190.5 ± 13.9°C (375 ± 25°F) provided the part is baked for 23 hours minimum after completion of the cadmium plating.

5.2.2 Plating sequence. If chromium and cadmium are used in combination, the chromium shall be deposited first. When chrome plating is to be followed by cadmium plating, the 23 hours minimum bake period can be replaced by a four (4) hour bake period at 190.5 ± 13.9°C (375 ± 25°F) provided the part is baked for 23 hours minimum after completion of the cadmium plating.

5.2.3 Storage of parts. Storage of parts between stress relief and cleaning shall be controlled to prevent contact with water or other corrosive materials. Parts shall be stored to permit free circulation of air around the parts.

5.2.4 Handling of parts. After the parts have been cleaned, they shall be handled in such a manner (white gloves, etc.) that will ensure a minimum of contamination.

5.2.5 Masking. Sections or areas of a part that are not to be plated shall be masked off. Plugging and masking materials, which do not contaminate the plating bath, shall be used. Masking shall be performed at the most convenient step prior to plating.

5.2.6 Racking. Sufficient contact area and pressure shall be provided to carry the current without overheating. Racking should be performed at the most convenient step prior to plating.

5.3 Plating procedures. The cadmium plating procedure shall be as described below:

5.3.1 Step Number 1. Parts shall be degreased as necessary. No minimum elapsed time requirement shall apply between this operation and the cleaning operation of Step Number 2.

5.3.2 Step Number 2. All parts shall be cleaned by dry blasting using 80 – 320 grit aluminum oxide (AI_2O_3), silicon dioxide (SiO_2) or garnet per MIL-STD-1504. Elapsed time between completion of cleaning and Step Number 3 shall not exceed four (4) hours.

5.3.3 Step Number 3. Rinse parts in cold water. A cyanide holding tank can be used to hold parts if the plating tank is full.

5.3.4 Step Number 4. Cadmium plate at 5.4 to 7.5 amperes/sq decimeters (50 – 70 amperes/sq foot) in the following solution:

Cadmium metal	21 – 30 g/l (2.8 – 4.0 oz/gal)
Total cyanide	90 – 135 g/l (12 – 18 oz/gal)
Sodium hydroxide	11.2 – 30 g/l (1.5 – 4 oz/gal)
Sodium carbonate	0 – 60 g/l (0 – 8 oz/gal)
Ratio NaCN/Cd	3.5/1 to 6.5/1

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Temperature	15.5 – 26.6°C (60 – 80°F) Total permissible iron 700 parts/million

5.3.5 Step Number 5. Rinse parts in cold water

5.3.6 Step Number 6. Rinse parts in hot water and blow dry with compressed air. Elapsed time between completion of plating and start of baking, Step Number 7, shall not exceed four (4) hours.

5.3.7 Step Number 7. Bake all parts heat treated above 1.24 X 10^9Pa (180,000 psi) for 23 hours minimum, at 190.5 ± 13.9°C (375 ± 25°F).

NOTE: Refer to paragraphs 5.4.1, 5.4.2, and 5.4.3 if post plating treatment is required. If post plating treatment is not required, refer paragraph 5.5.

5.4 Post plating treatment.

5.4.1 Types of plating. Refer to applicable directives for type of plating. If the type of plating is not specified, the part shall be given the Type II treatment.

Type I No post plating treatment required (See: paragraph 5.5)

Type II (See: paragraph 5.4.2)

Type III (See paragraph 5.4.3)

5.4.2 Type II.

a. Immerse in the Type II chromating solution for 15 to 30 seconds. Cadmium plated surfaces intended for conversion to Type II that have become passive as a result of the baking operation may be reactivated by a brief immersion in dilute acid.

b. Rinse thoroughly in clean water. An additional rinse in warm water (60°C {140°F} maximum) may be used to facilitate drying.

c. Inspect per paragraph 5.5.

5.4.3 Type III.

a. Immerse parts in Type III (reference SAE-AMS-QQ-P-416) phosphate solution for 15 to 20 seconds.

b. Rinse thoroughly in clean water. An additional rinse in warm water (60°C {140°F} maximum) may be used to facilitate drying.

c. See paragraph 5.5.

5.5 Inspection. Inspection shall be in accordance with the production control inspection and tests in SAE-AMS-QQ-P-416 and this standard.

5.6 Process control. Solutions and equipment used in the plating process shall be checked periodically and maintained in accordance with the requirements of this process standard.

5.7 Qualification embrittlement test. The processor shall demonstrate their ability to provide cadmium plate, which meets the requirements of paragraph 4.4 of this standard, as follows:

a. Four notched steel specimens per ASTM F 519, Type 1a.1 or 1a.2 shall be prepared.

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b. The specimens shall be prepared for and plated in accordance with all of the requirements of this standard. During plating the specimens shall be mounted symmetrically on a rack by themselves. All areas of the rack except the contact area shall be coated with a suitable maskant. An ammeter having a sensitivity of 0.5 amperes or better shall be connected between the specimen rack and the cathode. The specimens shall be plated at 5.4 A/dm^2 (50 $amps/ft^2$) to a minimum thickness of 0.020mm (0.0008 inch). Thickness shall be measured on the major diameter of the specimen, it is understood that the coating will be thinner in the notch area. The specimens shall be baked for 23 hours at 190.5 ± 13.9°C (375 ± 25°F) within four (4) hours of removal from the bath. After baking, the specimens shall be given the same type II treatment used for production components (to include reactivation with dilute acid, if used for production).

c. The specimens shall be subjected to 200 hours of static loading at 75 percent of the ultimate notched tensile strength. The test shall be considered passed if all four (4) specimens meet the requirements of paragraph 4.4.

d. Upon successful completion of the static load test, one of the notched tensile specimens shall be sectioned across the notch parallel to the axis of the specimen. Photomicrographs shall be taken of the notched area and examined for complete coverage of the notch with plating (Use 80 – 100 X magnifications).

e. A complete analysis report of the plating bath with the qualification test results shall be submitted, as required by, the procuring activity.

5.8 Re-qualification. Any significant change to the solution shall require re- qualification. Examples of significant changes are, if more than 20% of the solution is discarded and re-made, if the solution is dumped and a new solution made up, or if the solution is transferred out the processing tank. Re-qualification shall be accomplished by either of the following methods:

a. Re-qualify using the sustained notched tensile method in the above paragraphs 5.7.a through 5.7.d.

b. Re-qualify using the hydrogen detection instrument method in paragraph 5.9.1 and running the test in duplicate.

5.9 Process control embrittlement acceptance test. The process control embrittlement acceptance test shall be one of the following two methods:

5.9.1 Method 1, Hydrogen detection instrument testing.

a. Hydrogen detection instrument testing shall be performed within the processor's facility by a certified operator according to the instrument manufacturer's instructions.

b. Hydrogen detection instrument testing shall be conducted at least twice weekly.

c. The lambda$_{pc}$ (λ_{pc}) as determined by the test procedure shall be less than or equal to 80. if the λ_{pc} is greater than 80 but less than or equal to 96, treat the solution and re-test. If the λ_{pc} is greater than 96 but less than or equal to 130, halt production, investigate and correct the problem, and re-qualify the solution per paragraph 5.7 If λ_{pc} greater than 130 halt production reject all parts plated since the last acceptance test, correct the problem and re-qualify the solution per paragraph 5.7.

d. If the embrittlement test has not been performed in the 30 days preceding the processing of the material batch the bath must be re-qualified in accordance with paragraph 5.7.

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5.9.2 Method 2, Notched tensile tests.

a. Two standard specimens of the type noted in paragraph 5.7 shall be processed in conjunction with the plating of items in accordance with all of the requirements of this standard. The specimens shall be subjected to a sustained load test of 75 percent of the ultimate notched tensile strength of the material for 200 hours minimum and shall meet the requirements of paragraph 4.4. Failure of any one of the specimens shall constitute failure of the test and production shall cease until the cause of the failure is determined and the bath is re-qualified. Acceptance of items completed after the last successfully completed acceptance test shall be withheld until the extent and cause of the failure has been determined.

b. The test for embrittlement may be conducted as often as deemed necessary and shall be tested at least once monthly. If the embrittlement test has not been performed in the 30 days preceding the processing of the material batch the bath must be re-qualified in accordance with paragraph 5.7.

5.10 Safety and health. This document specifies the use of certain materials, which have been listed in 29 CFR 1910 (OSHA Standards) as "Toxic and Hazardous Substances." Personnel exposure to these materials during the process must be limited to the values specified in the applicable portions of the OSHA Standard 1910.1000.

6. NOTES:
This section contains information of a general or explanatory nature that may be helpful but is not mandatory.

6.1 Intended use. This standard provides guidance on the procedures and processes for cadmium plating, low embrittlement, and electro-deposition for the repair of existing or procurement of new parts on the landing gear for all military aircraft.

6.2 Alternatives. For use on high strength steels, low hydrogen embrittlement zinc nickel plating (LHE Zn-Ni) has been proven to be equivalent and in many cases superior to cadmium plating. The cognizant engineering authority should consider LHE Zn-Ni plating as an alternative to cadmium plating. MIL-DTL-32648 LHE Zn-Ni with the same Type and Class as specified for the original cadmium coating is a recommended substitute for cadmium plating on high strength steels.

6.3 Subject term (key word) listings:

Method Procedure

Sequence Type

6.4 Change notations. The margins of this standard are marked with vertical lines to indicate modifications generated by this change. This was done as a convenience only and the Government assumes no liability whatsoever for an inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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